

Homomorphism and Isomorphism.

Homomorphism : Let $(G,*)$ and (H,Δ) be any two groups. A mapping $f: G \rightarrow H$ is said to be a homomorphism if $f(a*b) = f(a)\Delta f(b)$, for any $a, b \in G$.

Ex1: Let $g: (Z, +) \rightarrow (Z, +)$ defined by $g(x) = 2x$ for all $x \in Z$

Then for $x, y \in Z$

$$g(x + y) = 2(x + y) = 2x + 2y = g(x) + g(y)$$

$\therefore g$ is a homomorphism.

Ex2: Let $(R^+, \cdot)$ be a group of positive real numbers under multiplication. Let $(R, +)$ be the group of all real numbers under addition.

Define $f: R^+ \rightarrow R$ by $f(x) = \log_{10}^x$. The binary operation for the groups R^+ and R are $\cdot$ and $+$ respectively.

To prove: f is a homomorphism.

i.e. to show that $f(x \cdot y) = f(x) + f(y)$ for all $x, y \in R^+$

Now,

$$f(x \cdot y) = \log_{10}^{xy} = \log_{10}^x + \log_{10}^y = f(x) + f(y) .$$

$\therefore f$ is a homomorphism.

Theorem: Consider the groups $(G_1, *)$ and $(G_2, \oplus)$ with identity elements e_1 and e_2 respectively. If $f : G_1 \rightarrow G_2$ is a group homomorphism, then prove that

a) $f(e_1) = e_2$

b) $f(a^{-1}) = [f(a)]^{-1}$

Proof: a) We have in G_2 ,

$$\begin{aligned} e_2 \oplus f(e_1) &= f(e_1) && \text{(since, } e_2 \text{ is identity in } G_2 \text{)} \\ &= f(e_1 * e_1) && \text{(since, } e_1 \text{ is identity in } G_1 \text{)} \\ &= f(e_1) \oplus f(e_1) && \text{(since } f \text{ is a homomorphism)} \\ e_2 &= f(e_1) && \text{(By right cancellation law)} \end{aligned}$$

b) For any $a \in G_1$, we have

$$f(a) \oplus f(a^{-1}) = f(a * a^{-1}) = f(e_1) = e_2$$

$$\text{and } f(a^{-1}) \oplus f(a) = f(a^{-1} * a) = f(e_1) = e_2$$

$$\therefore f(a^{-1}) \text{ is the inverse of } f(a) \text{ in } G_2 \text{ i.e., } [f(a)]^{-1} = f(a^{-1})$$

Theorem 2: Let f be a homomorphism from G to G' . Then $f(G)$ (the homomorphic image of G into G') is a subgroup of G'

Proof:

Let $f(G) = \{f(x) \mid x \in G\}$

Clearly $f(G)$ is a non-empty subset of G' . Now for any $a', b' \in f(G)$, we have $f(a) = a', f(b) = b'$, for $a, b \in G$.

$$\begin{aligned} a' * (b')^{-1} &= f(a) * f[b]^{-1} \\ &= f(a) * f(b^{-1}) \\ &= f(a * b^{-1}) \end{aligned}$$

But $a * b^{-1} \in G \Rightarrow f(a * b^{-1}) \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$

i.e., For $a', b' \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$

$\Rightarrow f(G)$ is a subgroup of G' .